assignment_rotation1

This is a preview of the published version of the quiz

Started: Apr 29 at 8:05pm

Quiz Instructions

Question 1 12 pts

When a rigid body rotates about a fixed axis, all the points in the body have the same

- tangential acceleration.
- linear displacement.
- tangential speed.
- centripetal acceleration.
- angular acceleration.

Question 2 12 pts

How long does it take for a rotating object to speed up from 15.0 rad/s to 33.3 rad/s if it has a uniform angular acceleration of 3.45 rad/s^2?

- 5.30 s
- 10.6 s
- 4.35 s
- 63.1 s
- 9.57 s

Question 3 12 pts

A child is riding a merry-go-round that has an instantaneous angular speed of 1.25 rad/s and an angular acceleration of 0.745 rad/s^2. The child is standing 4.65 m from the center of the merry-go-round. What is the magnitude of the linear acceleration of the child?

- $8.05\ m/s^2$
- $2.58\ m/s^2$
- $4.10\ m/s^2$
- $7.27\ m/s^2$
- $3.46\ m/s^2$

Question 4 12 pts

A wheel accelerates with a constant angular acceleration of 4.5 rad/s^2 from an initial angular speed of 1.0 rad/s. (a) Through what angle does the wheel turn in the first 2.0 s, and (b) what is its angular speed at that time?

- (a) 11 rad (b) 10 rad/s
- (a) 11 rad/s (b) 20 rad/s
- (a) 20 rad (b) 10 rad/s
- (a) 10rad (b) 20 rad/s

Question 5 12 pts

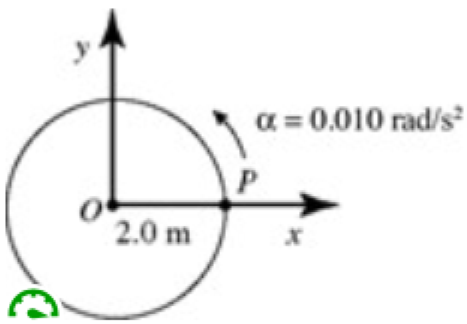
A scooter has wheels with a diameter of 120 mm. What is the angular speed of the wheels when the scooter is moving forward at 6.00 m/s?

hint: just the relationship linear speed and angular speed. 1 mm = 0.001m. you can check online how to convert revolution per minute to rad/s. 1 revolution is 2pi. 1 imin is 60s.

- 47.7 rpm
- 100 rpm
- 50.0 rpm
- 72.0 rpm
- 955 rpm

Question 6 12 pts

In the figure, point *P* is at rest when it is on the *x*-axis. The linear speed of point *P* when it reaches the *y*-axis is closest to

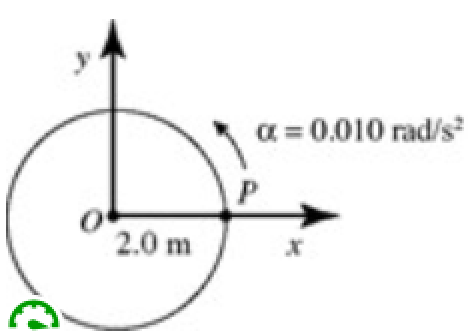


hint: use kinematics. the aangular displacement is pi/2 . To find the linear speed multiply the angular speed by R.

- 0.35 m/s.
- 0.71 m/s.
- 0.24 m/s.
- 0.18 m/s.
- 0.49 m/s.

Question 7 12 pts

In the figure, point *P* is at rest when it is on the *x*-axis. The time *t*, when *P* returns to the original position on the *x*-axis, is closest to



hint: use kinematics equation. The angular displacement is 2pi

- 35 s.
- 50 s.
- 25 s.
- 13 s.
- 18 s.

Question 8 12 pts

A 4.50-kg wheel that is 34.5 cm in diameter rotates through an angle of 13.8 rad as it slows down uniformly from 22.0 rad/s to 13.5 rad/s. What is the magnitude of the angular acceleration of the wheel?

- 0.616 rad/s^2
- 5.45 rad/s^2
- 10.9 rad/s^2
- 111 rad/s^2
- 22.5 rad/s^2

Question 9 12 pts

A bicycle has wheels that are 60 cm in diameter. What is the angular speed of these wheels when it is moving at 4.0 m/s?

- 7.6 rad/s
- 13 rad/s
- 0.36 rad/s
- 1.2 rad/s
- 4.8 rad/s

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